

Some Convex Geometry Associated with Erdos Problem No. 1120

This note is just an amateur project: errors are likely & corrections are welcome!

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The surprisingly elementary statement of Erdos Problem No. 1120 is the following:

(Statement of 1120) Let $f(z) \in \mathbb{C}[z]$ be a monic polynomial of degree n whose roots satisfy $|z| \leq 1$, and let $E_f = \{z : |f(z)| \leq 1\}$. What is the length of the shortest path contained in E_f from the origin $z = 0$ to the unit circle $|z| = 1$ as a function of n ?

According to the corresponding entry <https://www.erdosproblems.com/1120> in the Erdos database, it is known that such a path always exists [1] and that Erdos conjectured “presumably this tends to infinity with n , but not too fast”.

Our goal here is to use a bit of convex geometry to try to argue that for M sufficiently large, there must exist angles $\theta_1, \dots, \theta_M \in [0, \pi]$ such that $f(z) = \prod_{1 \leq j \leq M} (z - e^{i\theta_j})(z - e^{-i\theta_j}) = \prod_{1 \leq j \leq M} (z^2 + 1 - 2z \cos(\theta_j))$ has magnitude > 1 on concentric arcs of radii $0 < \rho_1 < \rho_2 < \dots < \rho_k < 1$ with small gaps on the side:

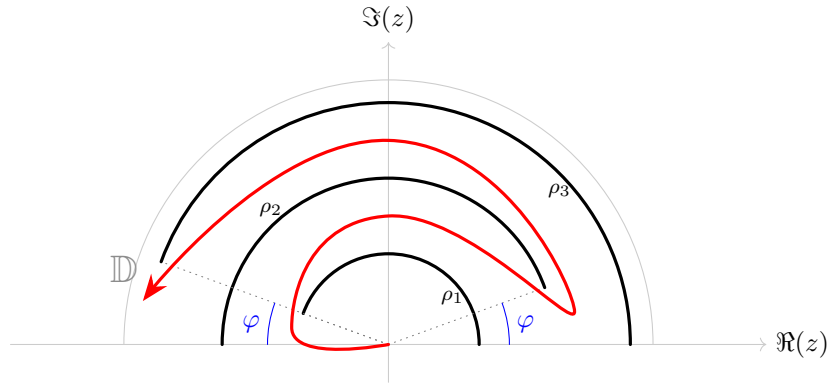


Figure 1: A circuitous path avoiding prohibited rings at radii ρ_1, ρ_2, ρ_3 through a gap angle of φ .

The idea is that since a path from $z = 0$ to $|z| = 1$ must exist, these ring barriers force the trajectory to take a winding, circuitous route whose length can be made arbitrarily long with k , and hence with $n = \deg f = 2M$. Hopefully this adds some credence to Erdos’ presumption on the unbounded growth for large n .

Clearly the choice of f as having roots strictly *on* the unit circle and coming in complex conjugate *pairs* is an imposed assumption: driven largely by its simplicity, numerical stability, and clear guarantee on the restriction of the roots only to the disk $\mathbb{D}$. More on this in §3.

Chebyshev Relaxation

The well-known formula for the generating function of the Chebyshev Polynomials, ie $T_n(\cos(t)) = \cos(nt)$, provides a delightful series expansion for the quantity $\ln(1-2z \cos \theta + z^2) = -2 \sum_{n=1}^{\infty} \cos(n\theta) \frac{z^n}{n}$ convergent for $|z| < 1$. This strongly motivates writing;

$$\ln f(z) = -2 \sum_{n=1}^{\infty} \frac{z^n}{n} \underbrace{\sum_{j=1}^M \cos(n\theta_j)}_{Z_n} \implies \ln |f(re^{i\phi})| = \Re \left(-2 \sum_{n=1}^{\infty} \frac{r^n e^{i\phi n}}{n} Z_n \right) = -2 \sum_{n=1}^{\infty} \frac{r^n}{n} \underbrace{\cos(n\phi)}_{T_n(\cos \phi)} Z_n$$

And so upon calling $g(r, w) = -2 \sum_{n \geq 1} \frac{r^n}{n} T_n(w) Z_n$ for $w \in [-1, 1], 0 \leq r < 1$, we obtain a *linear functional* in the Z_n variables whose positivity/negativity directly encodes whether the point $z = r(w + i\sqrt{1-w^2})$ lies within E_f . Treating the Z_n 's as truly free quantities gives us a very nice computational foothold:

(Chebyshev Relaxation) Let $D \subseteq \mathbb{D}$ define the region in (r, w) -coordinates we wish to prohibit path access. Must there exist an N sufficiently large, a vector $Z \in \mathbb{R}^N$, and a positive $\delta > 0$ such that:

$$\begin{bmatrix} r^1 T_1(w) \\ \vdots \\ r^N T_N(w) \end{bmatrix} \cdot Z \geq \delta, \quad \forall (r, w) \in D? \quad (1)$$

And so by-relabeling $Z_n = -n \frac{[Z]_n}{2}$ for $1 \leq n \leq N$ and $Z_n = 0$ for $n > N$, we thus obtain that $g(r, w) \geq \delta$ is indeed strictly positive on the desired domain D , indicating that the same might be true for f .

There are many nice properties about 1: for starters, by picking some large, finite collection of points (r_i, w_i) for D , one receives a Linear Program in $c_i = [r_i^1 T_1(w_i), \dots, r_i^N T_N(w_i)]$;

$$\max_{Z, \delta} \delta \quad \begin{bmatrix} -c_1 - \\ \vdots \\ -c_k - \end{bmatrix} Z \geq \delta \quad \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \quad (2)$$

And much computational time and energy was placed into running large and precise instances of such an LP, although considerable difficulty was encountered given the tight tolerances involved with determining whether the optimum satisfied $\delta > 0$. Additional semidefinite constraints can be added to 2 to produce a Convex Program whose solution vectors Z are closer to being “realizable” in the sense of §2.

Importantly, the vector Z in 1 plays precisely the role of a *separating hyperplane*. Indeed, if we let $\mathcal{C} = \{[r^1 T_1(w), \dots, r^N T_N(w)] : (r, w) \in D\}$ be the set of all constraint vectors involved, then Z is a separating hyperplane between the two convex sets $\{0\}$ and $\text{Conv}(\mathcal{C})$ (the convex hull). The following classic result of Convex Geometry is what generates Z for us:

(Separating Hyperplane Theorem) Let A and B be two closed, convex subsets of $\mathbb{R}^N$ with $A \cap B = \emptyset$. If at least one of A, B is compact, then there exists a unit vector $v \in \mathbb{R}^N$ and constants $c_1 > c_2$ such that $a \cdot v > c_1$ and $b \cdot v < c_2$ for all $a \in A, b \in B$.

Trivially $B = \{0\}$ is compact and gives $a \cdot v > c_1 > c_2 > 0 \cdot v = 0$ and so $c_1 = \delta$ gives us the desired positivity bound. Ensuring $A = \text{Conv}(\mathcal{C})$ can be made closed and ensuring $0 \notin A$ is thus our next step.

Our construction involving the concentric rings 1 boils down to studying the following closed, con-

vex set:

$$\left\{ \sum_{1 \leq i \leq k} \lambda_i \int_{-1}^1 \begin{bmatrix} \rho_i^1 T_1(t) \\ \vdots \\ \rho_i^N T_N(t) \end{bmatrix} d\mu_i(t) \right\} \quad (3)$$

Which is taken as $\lambda_1, \dots, \lambda_\ell$ ranges over the standard simplex of non-negative convex weights and μ_i runs over the space of probability measures with support $[l_i, u_i]$: the set of w -values corresponding to the blocked segments of the ring ρ_i , for example something like $[\cos(\pi - \varphi), 1]$ to represent an arc stretching from $\arccos(1) = 0$ radians leaving a φ -radian gap on the LHS or $[-1, \cos(\varphi)]$ leaving a φ -radian gap on the RHS. That either $-1 < l_i$ or $u_i < 1$ is critical to ensuring that a path from $z = 0$ to $|z| = 1$ can exist, and indeed that there exists a closed interval $[l'_i, u'_i] \subseteq [-1, 1]$ disjoint from $[l_i, u_i]$ will be crucial in a later step.

For example, setting $\lambda_i = 1, \lambda_{\neq i} = 0$ and $d\mu_i(t) = \delta(t - w)$ to be the Dirac measure for some $w \in [l_i, u_i]$, we immediately see that 3 contains the vectors $[\rho_i^1 T_1(w), \dots, \rho_i^N T_N(w)]$ for each i and each valid w , and so using it is sufficient to enforce $g \geq \delta$ on each of the rings we desire to blockade.

Reference Request: 3 is pretty clearly convex and pretty clearly closed given that convex combination of probability measures gives a probability measure and that convergence of measures $\mu_n \rightarrow \mu$ is generally accompanied by the point-wise convergence of $\int f d\mu_n \rightarrow \int f d\mu$ in the space of continuous and bounded functions f . Instead of bashing the proof directly, it would be nice to know what this “closure of a convex hull” is generally regarded as in the literature.

To show that 0 is *not* contained in 3 (which is to say our desired separating hyperplane Z exists), take BWOC $\lambda_1, \dots, \lambda_k$ (non-negative and summing to 1) and probability measures $\mu_1, \dots, \mu_k$ such that:

$$\sum_{1 \leq i \leq k} \lambda_i \int_{-1}^1 \rho_i^n T_n(t) d\mu_i(t) = 0 \quad \forall n = 1, \dots, N \quad (4)$$

We assume such a selection is possible no matter how large N is, with the idea being that the resulting contradiction implies there must exist a sufficiently large N for which 0 is not contained in the above convex hull, implying a separating unit vector Z must exist for some sufficiently large N .

From here, the typical technique from the theory of measures is to study a generating function / transform of these Chebyshev moments, namely $\alpha_i(z) = \sum_{n \geq 1} z^n \left(\int_{-1}^1 T_n(t) d\mu_i(t) \right)$. $\alpha_i(z)$ obviously

converges for $|z| < 1$ since $|\alpha_i(z)| \leq \sum_{n \geq 1} |z|^n \left(\underbrace{\int_{-1}^1 |T_n(t)| d\mu_i(t)}_{\leq 1} \right) \leq \sum_{n \geq 1} |z|^n < \infty$, so we get to

work with a set of functions $\alpha_1, \dots, \alpha_k$ which are analytic in the interior of the unit disc. Call its coefficients $a_n^{(i)} = \int_{-1}^1 T_n(t) d\mu_i(t)$ for convenience. Much like how the *Stieltjes Transform* carries moments to an integral of a rational function $S_\mu(z) = \sum_{n \geq 0} z^{-1-n} \int t^n d\mu(t) = \int \frac{z^{-1}}{1-\frac{t}{z}} d\mu(t) = \int \frac{1}{z-t} d\mu(t)$, these α_i ’s are also tied to a measure transform:

$$\alpha_i(z) = \int_{-1}^1 \sum_{n \geq 1} z^n T_n(t) d\mu_i(t) = \int_{-1}^1 z \frac{t-z}{1-2tz+z^2} d\mu_i(t) \quad (5)$$

In terms of generating functions, 4 is encoded by the fact that $\mathcal{A}(z) = \sum_{i=1}^k \lambda_i \alpha_i(\rho_i z) = \sum_{n \geq 1} z^n \rho_i^n \sum_{i=1}^k \lambda_i a_n^{(i)} = \sum_{n > N} z^n \left(\sum_{i=1}^k \lambda_i \rho_i^n a_n^{(i)} \right)$ actually has an incredible amount of cancellation in the lower-order terms of its power series about the origin.

Note that since $\alpha_i(\rho_i z)$ is analytic on the open disk $|z| < \frac{1}{\rho_i}$ and since $\mathcal{A}(z)$ is a sum of a finite number of analytic functions, it too is analytic on the interior of the disk $|z| < \frac{1}{\rho}$, since this region is the intersection of all of the relevant radii of convergence. Here, $\rho = \rho_\ell$ is the largest/last radius for which $\lambda_\ell \neq 0$: after all, it is possible that the convex combination does not include contributions of measures all the

way up to μ_k . In this region, we get the pretty trivial bound $|\mathcal{A}(z)| \leq \sum_{n>N} |z|^n \sum_i^N \lambda_i \rho^n 1 = \frac{(\rho|z|)^{N+1}}{1-\rho|z|}$ which is nice since with N sufficiently large, $\mathcal{A}(z)$ can be made arbitrarily small.

This bound on $\mathcal{A}$ should imply that $\alpha_\ell(z)$ (recall ℓ is the largest ring for which $\lambda_\ell \neq 0$) actually converges all the way to the boundary $|z| = 1$ of the unit disk, since BWOc if $\alpha_\ell(z)$ had a singularity on the unit circle say for some $|z^*| = 1$ then $\mathcal{A}(z)$ would inherit the same singularity at $\frac{z^*}{\rho}$ at a distance of $\frac{1}{\rho}$ where we know that $\mathcal{A}$ is actually quite small (if λ_ℓ were 0 this singularity would not transfer to $\mathcal{A}$).

- Since $\alpha_1(\rho_1 z), \dots, \alpha_{\ell-1}(\rho_{\ell-1} z)$ are all analytic within the radius $|z| < \frac{1}{\rho_{\ell-1}}$ it follows that this singularity of $\alpha_\ell(\rho_\ell z)$ at a distance of $|z| = \frac{1}{\rho_\ell} < \frac{1}{\rho_{\ell-1}}$ simply cannot be canceled by any other singularity in the sum $\lambda_1 \alpha_1(\rho_1 z) + \dots \lambda_{\ell-1} \alpha_{\ell-1}(\rho_{\ell-1} z)$ since of course all of them are bounded in this region.
- So, let's approach the singularity of α_ℓ at z^* by $z' = \frac{z^*}{1+\epsilon}$ from within the interior unit disk, taking $\epsilon > 0$ sufficiently small so that $|\alpha_\ell(z')| \geq M$ blows up beyond any arbitrary bound M . Since $|\frac{z'}{\rho}| = \frac{1}{(1+\epsilon)\rho} < \frac{1}{\rho}$ we are justified in using the bound $|\mathcal{A}(\frac{z'}{\rho})| \leq \frac{(\rho|z'/\rho|)^{N+1}}{1-\rho|z'/\rho|} = \frac{(\frac{1}{1+\epsilon})^N}{1-\frac{1}{1+\epsilon}}$ which clearly can be made arbitrarily small with N .

- A quick reverse triangle inequality tells us these two are in contradiction:

$$|\mathcal{A}(z'/\rho)| = \left| \lambda_1 \alpha_1 \left(\frac{\rho_1}{\rho} z' \right) + \dots \lambda_{\ell-1} \alpha_{\ell-1} \left(\frac{\rho_{\ell-1}}{\rho} z' \right) + \lambda_\ell \alpha_\ell \left(\frac{\rho}{\rho} z' \right) \right| \geq \lambda_\ell \underbrace{|\alpha_\ell(z')|}_{\text{Unbounded}} - \underbrace{\left| \lambda_1 \alpha_1 \left(\frac{\rho_1}{\rho} z' \right) + \dots \lambda_{\ell-1} \alpha_{\ell-1} \left(\frac{\rho_{\ell-1}}{\rho} z' \right) \right|}_{\text{Analytic}}$$

Ultimately, we've established that $\alpha_\ell(z)$ converges at all points in $\mathbb{D}$ *including* on the unit circle $|z| = 1$. Dropping the ℓ subscripts for convenience, this let's us freely manipulate quantities like $\alpha(e^{i\phi})$ for any ϕ real, and so in particular we'll take $t \in (-1, 1)$ and restructure the sum back into Chebyshev polynomials:

$$\nu(t) = \frac{2}{\pi\sqrt{1-t^2}} \left(\frac{1}{2} + \frac{\alpha(e^{i \arccos t}) + \alpha(e^{-i \arccos t})}{2} \right) = \frac{2}{\pi\sqrt{1-t^2}} \left(\frac{1}{2} + \sum_{n \geq 1} \underbrace{\cos(n \arccos t)}_{T_n(t)} a_n \right)$$

Whose sole purpose is being a (not necessarily non-negative) continuous function on $(-1, 1)$ which satisfies $\int_{-1}^1 T_n(t) \nu(t) dt = a_n = \int_{-1}^1 T_n(t) d\mu(t)$ for $n \geq 1$ and $\int_{-1}^1 \nu(t) dt = 1 = \int_{-1}^1 d\mu(t)$: as seen by the usual $\int_{-1}^1 T_n(t) T_m(t) \frac{2}{\pi\sqrt{1-t^2}} dt = \delta_{m=n}$ orthogonality condition (except for at $n = m = 0$ where we get a $1/2$ factor).

Clearly, given that the monomial t^n can be expressed as a linear combination of the $1 = T_0(t), \dots, T_n(t)$ polynomials, this equality of Chebyshev measures between μ and ν tells us that the moments of μ and ν are equal for each $n \in \mathbb{N}_{\geq 0}$ as well. Nicely enough, this is sufficient to pin down $\mu = \nu$ given that both (one's a probability measure and the other is absolutely continuous) are Borel measures on $[-1, 1]$:

(Hausdorff Moment Problem [Uniqueness]) Let $[a, b]$ be a compact interval. A bounded Borel measure μ on $[a, b]$ is uniquely determined by its sequence of moments $m_n = \int_a^b t^n d\mu(t)$. That is, if two such measures μ and ν satisfy $\int t^n d\mu = \int t^n d\nu$ for all $n \geq 0$, then $\mu = \nu$.

We arrive at our desired contradiction by noting that μ , which we defined to have support only on $[l, u] \subsetneq [-1, 1]$, ie vanishing identically on the *gap* in the ring outside the specified angular section of either $[-1, \cos(\varphi)]$ or $[\cos(\pi - \varphi), 1]$, *cannot* be a real analytic function. Whereas, given that $T_n(t)$ expands to a polynomial in t , it's mostly clear that $\nu(t)$ *is* real analytic, being a convergent power-series in t centered at 0. By the identity theorem, a real-analytic function such as ν vanishing identically on some connected open neighborhood, such as the gap in support just mentioned, must be identically 0 on it's domain of analyticity, contradicting the fact that ν is non-zero: having integral 1.

Realization: Starr's Lemma

Equipped with some understanding of the role of Z_n , we are left with the question of how it can be realized, or approximately realized, by a legitimate polynomial f :

(Chebyshev Realization) A radius $r < 1$ is fixed and an arbitrary precision ϵ is given. For what vectors $Z \in \mathbb{R}^N$ does there exist an M sufficiently large and angles $\theta_1, \dots, \theta_M \in [0, \pi]$ such that the polynomial $f(z) = \prod_{j=1}^M (z^2 + 1 - 2z \cos(\theta_j))$ satisfies $|\frac{1}{M} \ln |f(z)| - g_Z(z)| \leq \epsilon$ for all $|z| \leq r$?

Clearly if such a realization is possible then we can take a g from 1 such that $g(z) \geq \delta > 0$ for all $z \in D$ and then successfully construct a polynomial f such that $D \cap E_f = \emptyset$: simply pick M large enough so that say $\epsilon < \frac{\delta}{2}$ and note that $\frac{1}{M} \ln |f(z)| \geq g(z) - \epsilon$ so $|f(z)|^M \geq \exp(\delta - \epsilon) > 1$ so we're done. The shortest path in E_f must avoid D and therefore take an extra long detour.

We might as well start with the simplest and most natural inequalities we can make by splitting up the relevant sums for f and g :

$$\frac{1}{2} \left| \frac{1}{M} \ln f(z) - 2 \sum_{n=1}^N \frac{z^n}{n} Z_n \right| = \left| \sum_{n=1}^{\infty} \frac{z^n}{n} \left(\frac{1}{M} \sum_{j=1}^M \cos(n\theta_j) \right) - \sum_{n=1}^N \frac{z^n}{n} Z_n \right| \leq \sum_{n=1}^N \frac{|z|^n}{n} \left| Z_n - \frac{1}{M} \sum_{j=1}^M \cos(n\theta_j) \right| + \sum_{n>N} \frac{|z|^n}{n} \frac{1}{M} \left| \sum_{j=1}^M \cos(n\theta_j) \right|$$

In a moment we will study how close the length- N vectors Z and $C_\theta = \frac{1}{M} [\sum_j \cos(1\theta_j), \dots, \sum_j \cos(N\theta_j)]$ can be in the 2-norm, so it's nice that the first sum can be upper bounded with Cauchy-Schwarz:

$$\leq \sqrt{\sum_{n=1}^N \frac{|z|^{2n}}{n^2}} \sqrt{\sum_{n=1}^N \left| Z_n - \frac{1}{M} \sum_{j=1}^M \cos(n\theta_j) \right|^2} + \frac{1}{N} \frac{r^{N+1}}{1-r} \leq \sqrt{\pi^2/6} \|Z - C_\theta\| + \frac{1}{N} \frac{r^{N+1}}{1-r} \quad (6)$$

Where the second sum was trivially bounded by $\leq \sum_{n>N} \frac{r^n}{N} \frac{1}{M} \sum_{j=1}^M 1 = \frac{1}{N} \frac{r^{N+1}}{1-r}$ which is quite crude given that $\sum_{j=1}^M \cos(n\theta_j)$ is likely going to oscillate significantly for n large and hence average close to 0 rather than 1.

Notice that the set of all reachable C_θ can be described with another gem of convex geometry: a rescaling of the M -fold *Minkowski Sum* of the curve $\mathcal{C}(t) = \{[\cos(1t), \dots, \cos(Nt)] : t \in [0, \pi]\}$ (we're picking M points along the curve, adding them together, and then rescaling down by a factor of M). Intuitively, the more copies of $\mathcal{C}$ that we average together, the more mixing and *smoothing* that occurs that might allow for a closer and closer approximation to Z . The formal theorem in this respect is the following:

(Starr's Corollary to Shapley-Folkman Theorem) Let $S_1, \dots, S_M$ be a family of compact subsets of $\mathbb{R}^N$ with $N \leq M$. Then, for any $x \in \text{Conv}(S_1 + \dots + S_M)$ there exists a $y \in S_1 + \dots + S_M$ such that $\|x - y\| \leq L\sqrt{N}$ where $L = \max(r(S_1), \dots, r(S_M))$ is the maximum "inner radius" of the sets. [2]

So, suppose that our $Z \in \mathbb{R}^N$ happens to lie within the convex hull of our trigonometric curve ie $Z \in \text{Conv}(\mathcal{C})$:

- Clearly $\mathcal{C}$ is compact since it is the image of $[0, \pi]$ under a continuous function.
- Recalling the usual interplay between Minkowski sums and convex hulls, note that $\text{Conv}(\mathcal{C} + \dots + \mathcal{C}) = M\text{Conv}(\mathcal{C})$ and so we'll set $x = MZ$ and extract the desired point $MC_\theta \in \mathcal{C} + \dots + \mathcal{C}$ such that $\|MZ - MC_\theta\| \leq L\sqrt{N}$.
- I have not defined what the inner radius $r(S_i)$ in Starr's Lemma is, but given that $L = \max(r(\mathcal{C}), \dots, r(\mathcal{C})) = r(\mathcal{C})$, it suffices to know that L is purely a function of N and the geometry of $\mathcal{C}$.
- This lack of dependence on M thus clearly allows us to approximate Z by an actually valid, realizable set of angles $\theta_1, \dots, \theta_M$ to an accuracy of $\|Z - C_\theta\| \leq \frac{L\sqrt{N}}{M}$ which tends $\rightarrow 0$ with M .

Even more interesting, we can actually control our luck by *rescaling* the vector Z : indeed since g is linear in Z such a re-scaling (except with scale factor 0 obviously) will preserve its regions of positivity while guaranteeing us the desirable $Z \in \text{Conv}(\mathcal{C})$ condition:

(Rescaling Lemma) Let $\mathcal{C} = \{[\cos(1t), \dots, \cos(Nt)] : t \in [0, \pi]\}$ be the trigonometric curve as before. For each N , the (2-norm) ball of radius $\frac{1}{5\sqrt{N}}$ is contained within the convex hull $\text{Conv}(\mathcal{C})$.

- It's pretty easy to see that the origin lies within $\text{Conv}(\mathcal{C})$, for example we can sample points $t = \frac{1}{2N}\pi, \frac{3}{2N}\pi \dots \frac{2N-1}{2N}\pi \in [0, \pi]$ and uniform weights $1/N$ and obtain the convex-combination $0 = \frac{1}{N} \sum_{j=1}^N \cos\left(\frac{2j-1}{2N}\pi k\right)$ for $k = 1, \dots, N$.
- Let S be a compact subset of $\mathbb{R}^N$ such that $0 \in \text{Conv}(S)$. To show that a ball of radius r is contained in $\text{Conv}(S)$, it suffices to show that for each unit-vector $\hat{a}$, there exists an element $s \in S$ such that $\hat{a} \cdot s \geq r$. Notice that we need not check this condition for all $s \in \text{Conv}(S)$, only the initial set of points S .
 - This is because BWOC, suppose there were an x in the ball centered at 0 of radius r such that $x \notin \text{Conv}(S)$. By the Separating Hyperplane Theorem (again, $\{x\}$ compact and the convex hull of the compact set S is closed), there exists a separating unit vector v and a constant c such that $v \cdot s < c$ for all $s \in \text{Conv}(S)$ while $v \cdot x > c$.
 - Since $v \cdot s < c$ holds for all $s \in S \subseteq \text{Conv}(S)$ it follows that it holds for the particular element s such that $r \leq v \cdot s$, which we know exists for each unit vector including v . So, $r \leq v \cdot s < c$ and so $v \cdot x > c \geq r \implies v \cdot x > r$ which is a contradiction by Cauchy Schwarz ie $r^2 < (v \cdot x)^2 \leq |v|^2 |x|^2 \leq 1r^2$.
- In our case, this sufficient condition translates to the following: for each set of coefficients $a_1, \dots, a_N$ such that $a_1^2 + \dots + a_N^2 = 1$ (each possible choice of unit vector), the *maximum value* of $P(t) = a_1 \cos(1t) + \dots + a_N \cos(Nt)$ on $[0, \pi]$ should exceed r . So, given constraints on the magnitudes of the coefficients of a trigonometric polynomial of degree N , how small can its maximum value be?
- Notice that $\int_0^\pi P(t) dt = 0$ since all of the cosines average to 0, which tells us the area of the positive part say $\int_0^\pi \max(0, P(t)) dt$ exactly equals the area over the negative part $\int_0^\pi \max(0, -P(t)) dt$.
 - So, the 1-norm $\pi \|P\|_1 = \int_0^\pi |P(t)| dt$ which is the sum of the two areas both above and below the horizontal axis must be exactly $2 \times$ the positive area $\int_0^\pi \max(0, P(t)) dt$.
 - By the mean value theorem, we know that the maximum value for $P(t)$ over $[0, \pi]$ must at least exceed the positive part $\frac{1}{\pi} \int_0^\pi \max(0, P(t)) dt = \frac{\|P\|_1}{2}$, so we now desire to lower-bound the L_1 norm of P .
- Next, note that $\|P\|_2^2 = \frac{1}{\pi} \int_0^\pi |P(t)|^2 dt \leq \|P\|_\infty \frac{1}{\pi} \int_0^\pi |P(t)| dt = \|P\|_\infty \|P\|_1$.
- The interesting part of the inequality comes from taking $P(t) = \frac{1}{2\pi} \int_{-\pi}^\pi P(y) D_N(t-y) dy$ where D_N is the *Dirichlet Kernel* of order N (ie $D_N(x) = \frac{\sin((n+\frac{1}{2})x)}{\sin(\frac{x}{2})}$) which replicates trigonometric series of degree N or less exactly.
 - So, $|P(t)| \leq \|D_N\|_\infty \frac{1}{2\pi} \int_{-\pi}^\pi |P(y)| dy = \|D_N\|_\infty \|P\|_1 = (2N+1) \|P\|_1$ which finally gives us the dependence on N . The maximum value of $|D_N(t)|$ occurs at 0.
- Since $\|P\|_\infty \leq (2N+1) \|P\|_1$ we can substitute this back to find $\|P\|_2^2 \leq (2N+1) \|P\|_1^2$ and so $\|P\|_1 \geq \frac{1}{\sqrt{2N+1}} \|P\|_2$ but of course by Parseval's equality of energy we know that $\frac{2}{\pi} \int_0^\pi P(t)^2 dt = \sum_k a_k^2 = 1$ so in fact $\|P\|_2 = \frac{1}{\sqrt{2}}$ by assumption on $a_1, \dots, a_N$ forming a unit vector.
- So putting everything together we know $\max_{t \in [0, \pi]} P(t) \geq \frac{1}{2\sqrt{2}\sqrt{2N+1}} \geq \frac{1}{5\sqrt{N}}$ as desired.

Reference Request: Given a trigonometric polynomial P of degree N , a *Nikolskii-type Inequality* is one of the form $\|P\|_p \leq C_{p,q} N^{1/q-1/p} \|P\|_q$ for some constant $C_{p,q}$ universal for all such trigonometric polynomials. Our derivation above was actually the $p = 2, q = 1$ case of this class of inequalities, ie $\|P\|_2 \leq C \underbrace{N^{1/1-1/2}}_{\sqrt{N}} \|P\|_1$. A general reference for the rescaling lemma or the optimal constant involved would be nice.

So, in summary our realization strategy is follows: Suppose we are given a unit vector $Z \in \mathbb{R}^N$ such that provably $g_Z(z) \geq \delta$ for all $z \in D$ with all points in D satisfying $|z| \leq r$ (for example, the prohibited rings at radii $\rho_1 < \dots \rho_k < 1$ would be enclosed by any r such that $\rho_k < r \leq 1$). We then rescale Z down to $\frac{Z}{c\sqrt{N}}$ (say $c = 5$), in turn weakening our bound to $g(z) \geq \frac{\delta}{c\sqrt{N}}$. Since $\frac{Z}{c\sqrt{N}}$ has sufficiently small magnitude to lie within the Convex Hull of $\mathcal{C}$, we appeal to Starr's Lemma to identify a C_θ with $\|Z - C_\theta\| \leq \frac{L\sqrt{N}}{M}$.

Our inequality 6 now reads as:

$$\left| \frac{1}{M} \ln |f(z)| - g_Z(z) \right| \leq 2\sqrt{\pi^2/6} L \frac{\sqrt{N}}{M} + \frac{2}{N} \frac{r^{N+1}}{1-r} = \epsilon(N)$$

Of course, we know that $|f(z)|^{1/M} \geq \exp\left(\frac{\delta}{c\sqrt{N}} - \epsilon(N)\right)$ and so we need $\frac{\delta}{c\sqrt{N}} - \epsilon(N) > 0$ so in particular what we want is to show that $\sqrt{N}\epsilon(N)$ can be made arbitrarily small by increasing N .

Interestingly, we need not even select a new vector Z to do so: simply *pad* the end of Z with additional 0s to increase its dimension (N) while leaving its norm and the function g_Z completely unchanged. Padding N until the $\frac{2\sqrt{N}}{N} \frac{r^{N+1}}{1-r}$ term is made as small as desired will dararstically increase the value of the $\sqrt{N} 2\sqrt{\pi^2/6} L \frac{\sqrt{N}}{M}$ term, but of course we can immediately take M sufficiently large to make its contribution as small as desired, so we're done (maybe)!

Future Edits:

- **Asymptotics:** It would be great if this general strategy could be used to get some weak lower bound on the shortest-path asymptotic as a function of $\deg f = n$. After all, a path which snakes through k rings has length somewhere around $\sim \pi k$ where we know there is a vague dependence of N and M (so in turn n) on k .
- **Computational Constructions:** A significant amount of time was devoted to running LPs and SDPs like shown in 2. I wish to talk about the results in a dedicated section.
- **Choice of f :** The choice of parameterizing f by complex-conjugate roots at angles $\theta_1, \dots \theta_M$ encodes some nice properties of the shape of E_f near $z = 0$: linear combinations of Z_1, Z_2 , etc. dictate this geometry.
- **Generalizations:** Where else can the convex relaxations of these types of constrained polynomial optimization problems lead?

Mathematica

Mathematica's real quantifier elimination engine provides

To motivate this

(Erdos 1120 - Ray Form) What is the smallest degree n for which there exists a monic polynomial f , all of whose roots lie in $\mathbb{D}$, with minimal origin-to-circle path exceeding 1?

With a bit of tinkering

Infinitely many values for c_1, c_2, c_3 roughly in this relative configuration will produce the exact same effect:

References

- [1] WK Hayman. Research problems in function theory: new problems. In *Proceedings of the Symposium on Complex Analysis (Univ. Kent, Canterbury, 1973)*, pages 155–180, 1974.
- [2] Ross M. Starr. Notes on the shapley–folkman theorem. <https://econweb.ucsd.edu/~rstarr/SFarticle.pdf>, 2026.